

Human-Computer Interaction

**Design-led Inquiry:
RtD + Wizard-of-Oz**

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Today's Agenda

- Research *for* vs. *through* design
- Research through Design (RtD) as inquiry
- Wizard-of-Oz: studying the not-yet-built system
- Think-aloud / usability as data-collection method
- Hands-on: *prototype the study before you build the system*

This week's readings

- **Required**
 - Zimmerman & Forlizzi, *Research Through Design in HCI* (2014)
 - Kelley (1984) – the origin of Wizard of Oz: simulate the system, study the interaction before it exists
- **Recommended** – LFH **Ch. 10, “Usability testing”** (263–298), for think-aloud
- **Optional** – Gaver, *Science and Design: Different Forms of*

Key concepts in design-based research

Research for design: research to inform the design of a product or service.

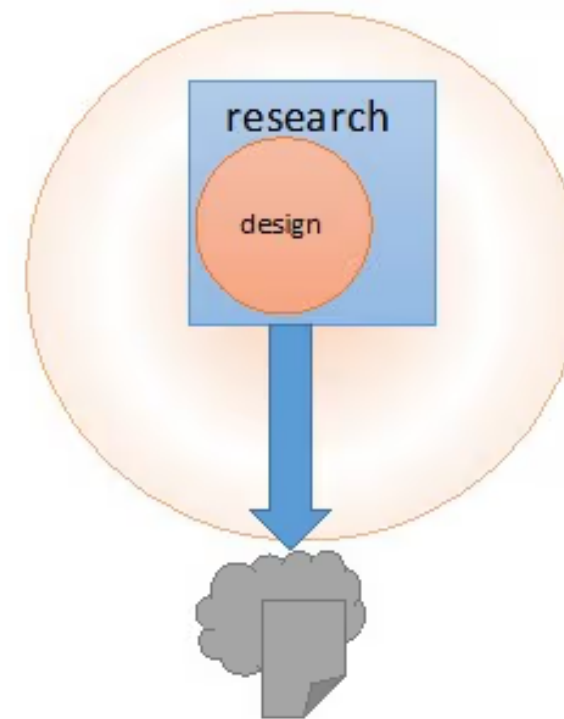
research → design

Research through design: design carried out to create knowledge about phenomena.

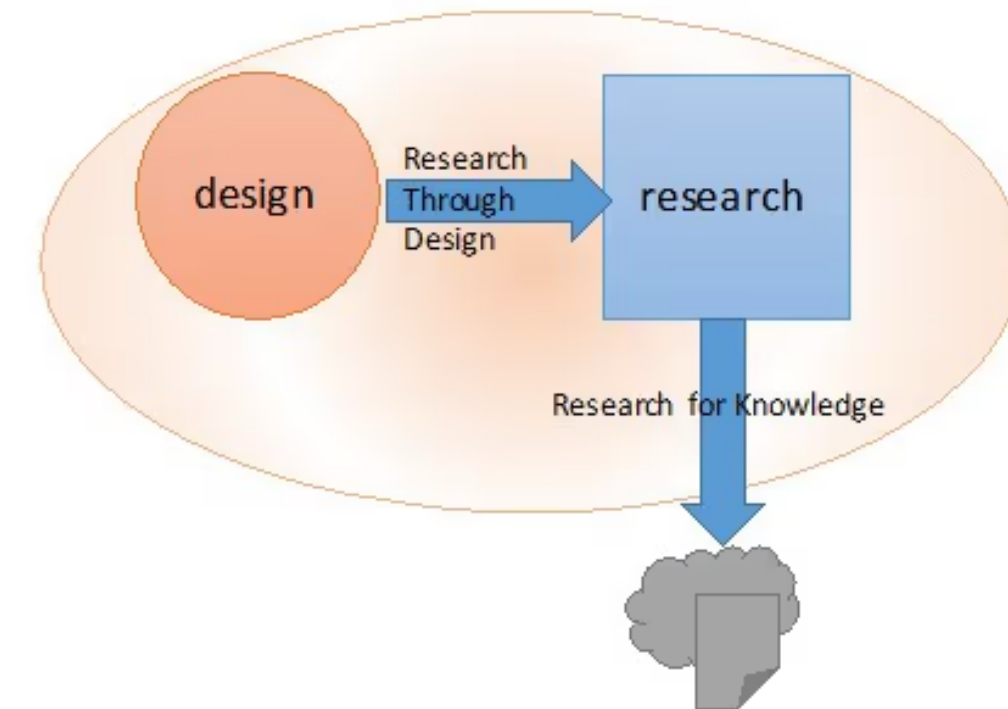
design ≈ research

How should we think about design and research?¹

design as part of research

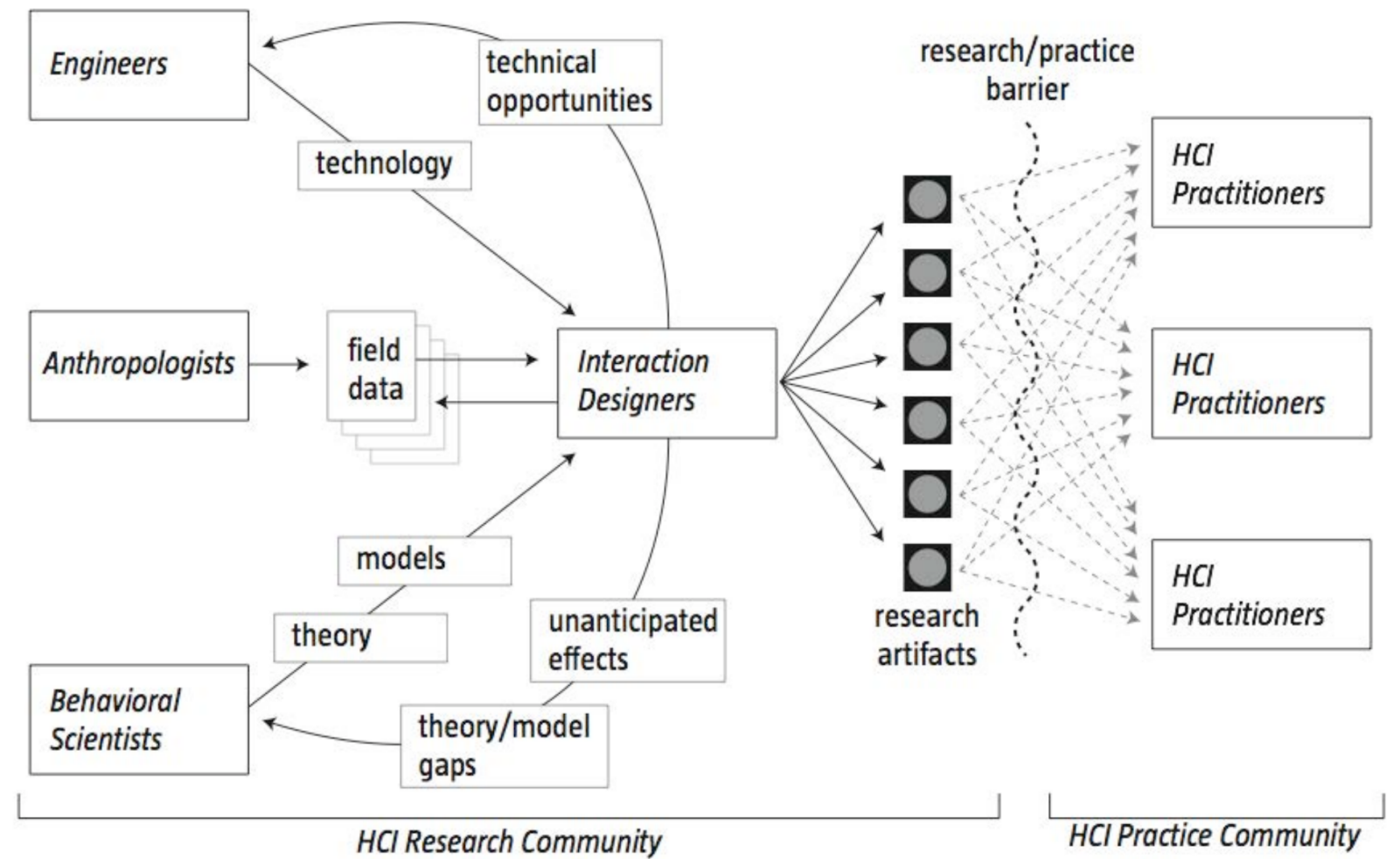


designerly ways of doing research



¹ Stappers & Giaccardi (2014). Research Through Design. *The Encyclopedia of HCI*, 2nd ed.

The RtD model²



²Zimmerman, Forlizzi & Evenson (2007). Research Through Design as a Method for Interaction Design Research in HCI. *CHI 2007*.

When to choose Research through Design

Consider RtD if:

- Your question is about *possibilities* ("what could be?") rather than existing phenomena
- Existing theory is nascent or fragmented
- You want to generate concepts, frameworks, or exemplars rather than validate measures
- Your contribution is *artifacts-as-arguments* (Zimmerman & Forlizzi)

When *not* to choose RtD

RtD may not fit if:

- The field already has mature constructs to refine or validate
- Your question is about causal relationships or generalizable behavioral effects
- The audience expects quantitative measures of effect

Research questions in RtD

Often framed as:

- "How might we...?" (*design problem framing*)
- "What new interaction possibilities can we reveal by building...?"
- "What does this artifact teach us about people, technology, or contexts?"

Preview: the experimental RQ

What are the effects of X on Y under conditions Z?

Variables

Target population,
experimental context, etc.

Check any RQ against **FINER**: **F**easible · **I**nteresting · **N**ovel · **E**thical · **R**elevant⁷

⁷Farrugia et al. (2010). Research questions, hypotheses, and objectives.

Process of RtD

1. **Problem framing:** identify an open or ill-structured space
2. **Iterative design + prototyping:** artifacts that embody a perspective
3. **Deployment / evaluation in context:** observe interaction, reflect on outcomes
4. **Knowledge articulation:** translate artifacts into design knowledge

What counts as data in RtD

- Prototypes and artifacts
- Observations and reflections from deployments
- The designer's annotated process – sketches, iterations
- Critiques (academic, practitioner, user communities)

An RtD example³

Unremarkable AI – fitting intelligent decision support into clinical workflow

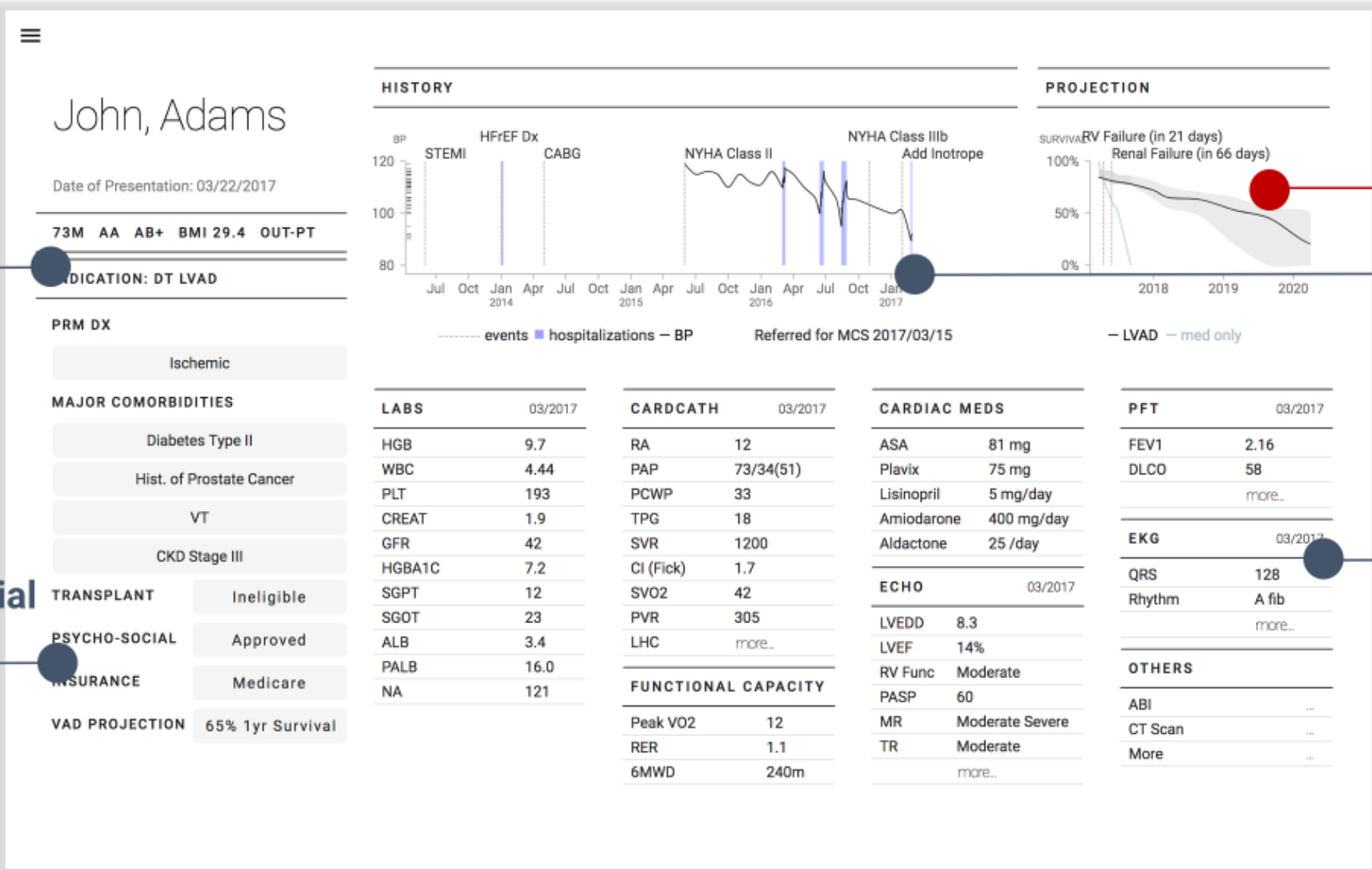
- **Problem framing:** why clinical DSTs fail in practice
- **Design:** predictions placed to be "*unremarkable*" within routines
- **Evaluation:** field assessment at 3 hospitals, on synthetic cases
- **Knowledge:** design AI to *blend into* existing work

³Yang, Steinfeld & Zimmerman (2019). Unremarkable AI. *CHI 2019*.

Where the prediction sat

Patient Demographics

Psychosocial and Financial
Evaluation Notes



DST Predictions

Patient History

Patient Conditions

*How do you study a system
you have not built yet?*

Wizard-of-Oz (WoZ)⁴

A human "wizard" **simulates the system's intelligence** while the participant interacts as if it were real.

- Study the *interaction* before the model exists
- Vary behavior faster than you could build it
- The design question moves upstream – *what should it do?* before *can we build it?*

⁴Kelley, J.F. (1984). An Iterative Design Methodology for User-Friendly Natural Language Office Information Applications. *ACM Transactions on Office Information Systems*, 2(1), 26–41. doi:10.1145/357417.357420 (Kelley coined the "Oz paradigm" in his 1983 Johns Hopkins dissertation; this is the archival paper.)

Designing a WoZ study

- **What the wizard simulates** – which capability is faked (recognition, reasoning, generation)
- **What the participant sees** – the interface, the latency, the persona
- **Where the seam is** – what is real vs. simulated, and how to keep it invisible
- **What you capture** – behavior, breakdowns, expectations against the illusion

WoZ in practice: Snackbot⁵

- **Problem framing:** designing for long-term HRI on campus
- **Design:** snack-delivery robot, iterated over 24 months
- **Evaluation:** *Wizard-of-Oz trials*, feasibility studies, early deployments
- **Knowledge:** documented holistic design process; sociable long-term robot guidelines

⁵ Lee et al. (2009). The Snackbot: Documenting the Design of a Robot for Long-Term HRI. *HRI 2009*.



Think-aloud as a data method⁶

In WoZ: what they expected the system to do – as data, not a to-do list.

- **Task:** a representative task they work through
- **Prompt:** "keep talking – what are you expecting?"
- **Capture:** breakdowns, expectations, confusion
- **Wizard's rule:** act only on what the real system could sense

⁶Lazar, Feng & Hochheiser (2017). *Research Methods in HCI*, Ch. 10, "Usability testing," 263–298.

Hands-on activity (30 min)

"Prototype the study before you build the system"

In your pod, take your team's project and design the *study* around a system you have **not** built:

- (12 min) *RtD / WoZ framing*. Specify the WoZ setup – what the wizard simulates, what the participant sees, where the seam is.
- (18 min) *Data-as-method pass*. Add a think-aloud / usability plan: what task, what you prompt participants to verbalize, what you capture as data. *Nothing to submit today*.

Grab the worksheet

- **One copy per team** – File → Make a copy
- Fill it in together; **nothing to submit today**
- Afterward, each of you copies it into your **own** Design-led / WoZ Study Plan

da.gd/5IQKv



Assignment – Design-led / WoZ Study Plan

Due: Fri **Oct 2, 9:30 am** – bring it to Methods 4.

1. **Study framing** – research *through* design, a systems contribution, or an empirical study of an existing system? Justify.
2. **WoZ / prototype-fidelity plan** – what is simulated vs. real; the seam; what the participant experiences.
3. **Think-aloud / usability data plan** – task(s), verbalization prompts, what is captured as data.
4. **Fit to RQ** – how the plan answers your current research

What's Next?

- **Monday – Seminar: Social Computing & CSCW**
 - **Read:** Grudin (1988) · Park et al., *Social Simulacra* (2022)
 - **If you can:** Schmidt & Bannon (1992) · Ackerman, the sociotechnical gap
 - Post your provocation slide
- **Project milestone (Wk 5-6):** Method design **begins**